The Price of Written Output

Human Labor, Language-Model Inference, and the Limits of a Token Comparison

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Abstract

What does a human-produced token cost? The question is simple, but its answer depends on which work, which output, and which time denominator are measured. This paper reanalyzes public data from a professional-writing experiment and combines the resulting task rates with U.S. occupational mean wages. A selected sample excluding reported AI use, comprising 149 grant writers, marketers, consultants, and human-resources professionals produced 70,254 reference-tokenizer tokens over 4,900 self-reported task minutes, or 860 tokens per task hour. The resulting synthetic wage cost is approximately $52,000 per million reference tokens if all paid hours are devoted to comparable tasks, $104,000 at one-half utilization, and $208,000 at one-quarter utilization. These are calibrations, not estimates of an average occupation's annual output. A snapshot of 12 official model tariffs reports $0.50–$50 per million billed output tokens, with selected hosted open-weight offerings at $1.20–$4.18. Input charges, reasoning, retries, different tokenizers, and review change the relevant cost of delivered work. An explicit stress scenario reaches $2,360 per million reference tokens before human review; review can dominate that charge. The evidence supports a large gap in the acquisition cost of text generation under stated assumptions. It does not identify an equivalent gap in quality-adjusted task costs, labor substitutability, or economic value.

Keywords: generative artificial intelligence; labor productivity; unit costs; professional writing; measurement.

JEL codes: J24, O33, D24.

Status: Independent working paper; not an NBER publication or a peer-reviewed article. Model prices were collected October 21, 2026, Chicago time, with selected October 22 checks. All computations and source records accompany the paper.

1. Introduction

Language models make text inexpensive to generate. Humans make text as one component of a much larger bundle of services: they investigate, select facts, interpret objectives, coordinate with others, exercise judgment, and accept responsibility for results. A comparison expressed in dollars per token can illuminate the cost of producing written material. It becomes misleading if it treats all of those services as interchangeable with the act of generating a string.

This paper starts with a deliberately narrow question: what wage expenditure would be associated with producing a specified quantity of professional written output at experimentally observed task rates, and how does that expenditure compare with advertised language-model inference prices? The numerator is cost and the denominator is output. Dividing output by wages instead produces tokens per dollar; both measures are reported where useful.

Three features distinguish the exercise from a comparison based on typing speeds. First, the human denominator comes from final text submitted in a published experiment, rather than keyboard activity or an assumed number of words per day. Second, the calculation distinguishes task time from paid employment time. Third, model charges are separated into billed input, visible output, reasoning, and subsequent review. Each source of evidence has its own date and population.

The main estimate combines four occupational groups in the Noy and Zhang (2023) replication data with occupation-specific wage proxies from the Bureau of Labor Statistics (BLS). These groups are selected because their experimental tasks yield countable prose and their titles permit an intelligible, if imperfect, wage crosswalk. The estimate is a synthetic unit cost for that selected task mix. It is not a new nationally representative survey of workers and does not identify the average American worker's output.

The principal finding is an orders-of-magnitude difference between calibrated human wage cost and the posted price of raw model generation. That difference is robust to substantial generation-side adjustments in the scenarios examined here. Its economic interpretation is narrower. Neither matched contemporary model performance nor employer acceptance of the experimental documents is observed. Review, fact-finding, organizational context, and the cost of errors are not free. A plausible employment conclusion therefore cannot be read directly from the price ratio.

2. Measurement and estimands

2.1 Human output and the time denominator

Let q_i denote the number of tokens in worker i's final submission under a fixed reference tokenizer, t_i the reported hours devoted to the task, and w_i an hourly wage proxy. Let u be the fraction of paid time devoted to comparable tasks. For an identical utilization assumption across the sample, the calibrated wage cost per million reference tokens is

$$H(u) = \frac{10^6 \sum_i w_i t_i}{u \sum_i q_i}.$$

An illustrative compensation multiplier b gives $H_b(u) = bH(u)$. The corresponding pooled task output rate is $v = \sum_i q_i / \sum_i t_i$, and paid-hour output under the utilization assumption is uv. The pooled wage entering the identity $H = 10^6 \bar{w}_t/(uv)$ is task-hour weighted, $\bar{w}_t = \sum_i w_i t_i / \sum_i t_i$. Neither an unweighted average of individual cost ratios nor an unweighted average of individual production speeds generally equals this estimand.

Task time includes reported brainstorming, drafting, editing, and other time. It is not merely typing time. The utilization adjustment represents paid time outside comparable completed tasks, such as unrelated duties, coordination, and leave. Time already included in the task report must not be deducted a second time. No utilization fraction is observed in the experiment. Values of 100%, 50%, and 25% are therefore explicit scenarios, with 100% serving as a full-utilization benchmark rather than a claim about real working days.

The measured output is a submitted document that passes the study's basic response screen. That is weaker than employer-accepted output meeting a specified factual, legal, or commercial standard. Accordingly, the empirical calculation uses the phrase *submitted reference tokens*. In a future matched-quality study, the denominator should be replaced by accepted output, with rejected work and revisions included in the cost numerator.

2.2 Model tariffs and delivered output

Providers bill their own token units. Let p_I and p_O be dollars per million native input and output tokens, r the native input-to-visible-output ratio, and h additional billed reasoning tokens per visible output token. If reasoning is billed at the output rate, the tariff per million native visible output tokens is

$$M_N(r, h) = p_O(1 + h) + p_I r.$$

This equation adds reasoning only when it is additional to the visible answer denominator. It must not be added again to an output count that already includes it. Some systems reason by default, so $h = 0$ is an accounting benchmark, not a demonstrated operating configuration for every model.

Let τ be native visible output tokens per reference output token, and let a be the probability that a complete, equally costly attempt is accepted. Under independent retries of equal expected cost,

$$M_R(r, h, \tau, a) = \frac{\tau M_N(r, h)}{a}.$$

Here $1/a$ is the expected attempt multiplier. The formula is not a general model of partial salvage, variable-length retries, or correlated failure. Those processes require actual attempt records. None of r, h, τ, or a is measured for a matched model execution in this paper. Tariff scenarios that set $\tau = 1$ should not be mistaken for measured tokenizer equivalence.

Finally, if review and other human handling require m paid minutes per 1,000 reference-token document at hourly cost c_R, the model-assisted cost per such document is

$$C_{1000} = M_R/1000 + c_R m/60 + K_{1000},$$

where K_{1000} collects other costs such as integration, information acquisition, and expected errors. Review minutes must cover all attempts and required follow-up, not just the final successful answer. Setting $K_{1000} = 0$ isolates an accounting comparison; it is not evidence that these costs vanish.

3. Data and sample construction

3.1 Professional-writing microdata

Noy and Zhang (2023) recruited 453 experienced, college-educated professionals through Prolific. The experiment ran from January 27 to February 21, 2023 and studied GPT-3.5 access. Participants completed two occupation-specific tasks, such as press releases, short reports, analysis plans, and delicate emails. The treatment group received ChatGPT registration instructions between tasks. The study used linear, convex, and fixed-time incentive arms. Its author-hosted final-version manuscript and public OSF replication files are the basis of the present reanalysis (Noy and Zhang 2023a, 2023b).

The replication pipeline reproduces the authors' 453-person cleaned sample from 786 raw survey rows, using their completion, duplicate, invalid-response, quality-screen, and prior-grader rules. It then applies additional filters chosen for this measurement question. These are new descriptive sample restrictions, not an attempted replication of the authors' causal treatment estimates.

Table 1. Sample construction for the human-output calibration

Restriction	Remaining people
Authors first-task observations	453
Confirmed quality screen	449
Non-code occupations	401
Unconstrained linear/convex incentive arms	316
No reported first-task ChatGPT or other AI	285
Nonempty text and positive complete reported task time	278
Four occupations with explicit wage crosswalks	149

Source: Authors' public raw survey, original cleaning code, grading screens, and full-text files; this paper's additional restrictions. Counts are sequential. The retained narrow sample contains one original first-task observation per person.

The broad eligible sample has 278 participants. The main calibration further excludes managers, whose titles are too broad for a defensible single wage proxy, leaving 149 grant writers, marketers, consultants, and HR professionals. Data analysts are excluded from the prose sample because code, analysis plans, and written descriptions have different output meanings. Technical writers and

medical transcriptionists have observable BLS wages but no matched production-time evidence in these data; they are not given invented output rates.

First-task text is selected using the randomized A/B task order. The untruncated full-text file supplies the original submitted text, rather than the largest intermediate word-count snapshot or the optional later retry. Whitespace-delimited words and `tiktoken` 0.12.0 `cl100k_base` tokens are counted from that text. The time denominator is the sum of all four self-reported components, requiring complete positive time and nonempty text. This avoids mixing a partial time report with a complete output count.

The phrase *no reported AI use* is essential. The first-task ChatGPT question was conditional on later reported use. A missing response therefore does not establish non-use. In the narrow sample, 94 of 149 first-task AI-use responses are unobserved and 55 explicitly report no ChatGPT use. The other-AI question covers either task, so excluding it may select first-task observations on subsequent behavior. Sensitivities include explicitly reported first-task non-use, removal of large paste events, and removal of these AI exclusions. None certifies that every word was produced unaided. Geographical eligibility was not established from the archived material; the experimental sample must not be described as a representative U.S. population.

3.2 Wages and compensation

Wages come from the May 2025 OEWS national workbook, the most recent vintage displayed in the BLS tables index when collected. The paper uses the **mean**, not median, hourly wage. The four crosswalks are HR professional to Human Resources Specialists (13-1071), consultant to Management Analysts (13-1111), grant writer to Writers and Authors (27-3043), and marketer to Market Research Analysts and Marketing Specialists (13-1161). The consultant and grant-writer mappings are broad proxies, and none supplies the experiment participant's actual wage. Differences in seniority and specialization remain unobserved.

OEWS covers wage-and-salary jobs and excludes self-employed workers. It measures wages, not full employer cost; many annual equivalents use 2,080 hours. The calculation uses published hourly wages directly rather than treating an annual salary as payment for 2,080 continuous writing hours (BLS 2026a, 2026b). The wage reference date is later than the experiment. Thus this is a May 2025 wage calibration at 2023 experimental task rates, compared with October 2026 tariffs. It is not a same-period price index or an inflation-adjusted causal comparison.

For a separate employer-cost sensitivity, June 2026 ECEC reports private-industry total compensation of \$46.89 per hour worked, wages of \$32.82, and paid-leave costs of \$3.54. The conventional total-to-wage ratio is 1.4287. Because the utilization denominator here is paid time and may already include leave, mechanically applying that ratio risks counting paid leave twice. The illustrative nonleave-adjusted ratio used here is $46.89/(32.82 + 3.54) = 1.2896$. This remains a broad-population proxy with different definitions from OEWS, not an estimated occupation-specific benefit rate. There is no additional paid-leave deduction or multiplier (BLS 2026c).

3.3 Model selection and price collection

The model sample covers current vendor-described flagship, workhorse, and efficiency families from OpenAI, Anthropic, and Google, plus hosted open-weight offerings from DeepSeek and Mistral. "Frontier" is used as a vendor-catalog selection criterion, not as the result of an independent benchmark ranking. The sample is deliberately transparent rather than exhaustive: Qwen, Llama, Kimi, and additional hosting providers are not comprehensively surveyed.

Prices are synchronous, uncached, standard or default-route API tariffs in USD. Subscription fees, free credits, cached-input rates, batch discounts, and premium speed tiers are excluded from the base table. Applicable published introductory tariffs are included and identified. DeepSeek's peak tariff is the conservative base, with off-peak prices shown separately. Raw source responses, hashes, collection times, API identifiers, and billing conditions are retained in the source ledgers.

The original snapshot was collected October 22 at approximately 03:44–03:47 UTC, which was October 21 in Chicago. October 22 rechecks confirmed unchanged visible pricing tables for OpenAI, Google, DeepSeek, and Mistral. Anthropic's refresh returned HTTP 403; its earlier successfully retrieved official documentation remains the source. A retrieval timestamp establishes what was observed then, not a page's publication date (OpenAI 2026; Anthropic 2026; Google 2026; DeepSeek 2026; Mistral AI 2026).

4. Human-output results

The narrow sample produced 59,347 words and 70,254 reference tokens over 4,900 reported minutes. The ratio of totals is 727 words or 860 tokens per task hour. The observed aggregate conversion is 1.184 reference tokens per whitespace word; using a generic four-thirds conversion would overstate token output by about 13% for this particular corpus. The result illustrates why text should be tokenized directly when possible.

Table 2. Occupational task rates and synthetic wage costs

Group	N	Wage/hour	Tokens/task hour	Cost: u=1	Cost: u=0.5
HR professional	24	39.42	828	47,603	95,206
consultant	38	54.71	852	64,185	128,369
grant writer	59	41.39	861	48,076	96,151
marketer	28	43.03	899	47,850	95,700
Pooled calibration	149	44.77	860	52,040	104,079

Notes: Rates use final first-task text and reported task time. Wage proxies are May 2025 national means. Costs are USD per million `cl100k_base` reference tokens; wage-only. The pooled row weights compensation by task hours and output by actual submitted tokens. It is not an employment-weighted U.S. average.

The pooled task-hour-weighted wage proxy is $44.77. At full utilization, the calibrated cost is $52,040 per million reference tokens, or about $0.052 per token and 19.2 tokens per dollar. At half

utilization, the cost is $104,079 per million, or about 9.6 tokens per dollar. At quarter utilization it is $208,158 per million. Differences across the four groups are driven substantially by wage proxies: their observed task rates occupy a relatively narrow range of 828–899 reference tokens per hour.

The half-utilization scenario implies about 430 submitted reference tokens per paid hour, or 3,441 over eight paid hours. At 2,080 paid hours it would imply approximately 895,000 reference tokens per year. These are mechanical annualizations of a short-task rate under assumed utilization; no worker's annual production was measured. The annual figure should not be quoted as an observed average.

The separate nonleave-adjusted compensation sensitivity raises the pooled half-utilization cost to approximately $134,221 per million reference tokens. This includes the broad proxy for nonwage compensation described above. It still excludes business premises, equipment, recruiting, management, and other organizational costs that cannot be attributed reliably using these sources.

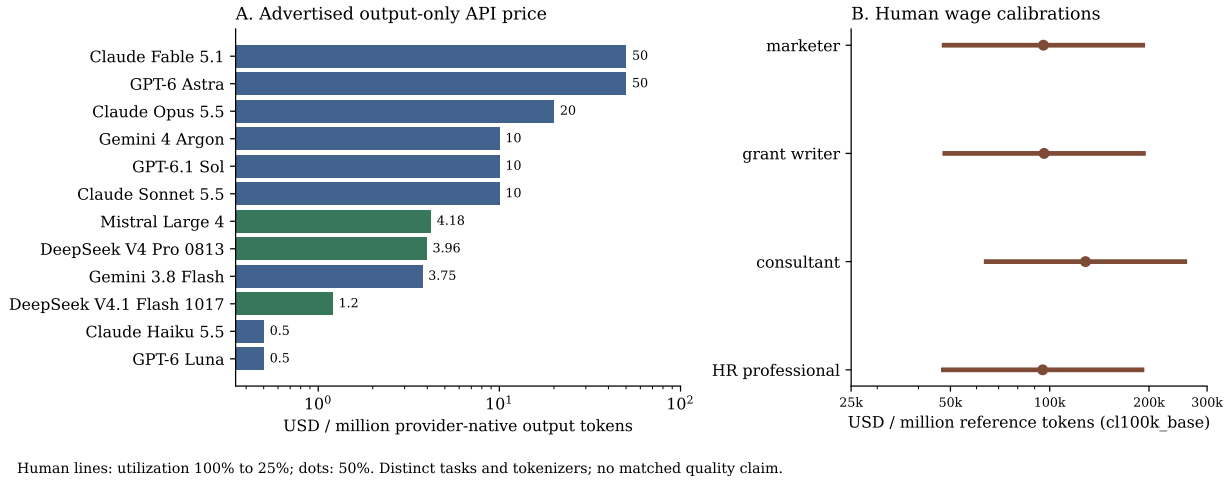


Figure 1: Model tariffs and calibrated human costs, shown in their distinct token units. The human line spans utilization from 100% to 25%; the dot marks 50%. These panels do not establish matched task quality.

5. Model prices and the arithmetic comparison

Table 3. Official model-price snapshot

Model	Weights	Input	Output	Input = output
GPT-6 Astra	Closed	10.00	50.00	60.00
GPT-6.1 Sol	Closed	2.00	10.00	12.00
GPT-6 Luna	Closed	0.10	0.50	0.60
Claude Fable 5.1	Closed	10.00	50.00	60.00
Claude Opus 5.5	Closed	4.00	20.00	24.00
Claude Sonnet 5.5	Closed	2.00	10.00	12.00
Claude Haiku 5.5	Closed	0.10	0.50	0.60
Gemini 4 Argon	Closed	2.00	10.00	12.00
Gemini 3.8 Flash	Closed	0.75	3.75	4.50
DeepSeek V4.1 Flash 1017	OW	0.30	1.20	1.50
DeepSeek V4 Pro 0813	OW	1.32	3.96	5.28
Mistral Large 4	OW	1.36	4.18	5.54

Notes: USD per million provider-native billed tokens. "Input = output" is the sum of input and output tariffs, with no additional reasoning or other charges. It is a tariff illustration, not a measured delivered-answer cost. OW denotes provider-described open weights; DeepSeek's cited licenses are MIT, while Mistral's exact license was not verified. Full API IDs and tariff conditions are in Appendix B and the machine-readable table.

Output-only prices span \$0.50–\$50 per million billed tokens in this selected sample. Hosted open-weight offerings span \$1.20–\$4.18 at the selected peak or standard tariffs. At equal input and visible-output counts, with no additional reasoning, the corresponding ranges are \$0.60–\$60 and \$1.50–\$5.54. Neither range is a price of self-hosting. Free access to model weights does not make inference free: utilization, hardware, operations, and redundancy would need separate measurement.

To put the orders of magnitude on a common accounting scale, temporarily assume one native visible token per reference token, no retries, and equal input/output counts. The full-utilization human calibration is then about 867 times the \$60 high-price model benchmark and about 86,733 times the \$0.60 low-price benchmark. At half utilization, these arithmetic ratios double. The denominators are not quality-matched outputs, so these are **conditional price ratios**, not productivity or intelligence multiples.

Tokenization is a material limitation, not merely a matter of rounding. Anthropic's documentation reports that its newer tokenizer produces approximately 30% more tokens for the same text than its preceding tokenizer. That comparison is not a mapping to `cl100k_base`, so no mechanical 30% correction is imposed here. Instead, a reference/native conversion factor is exposed in the sensitivity calculations. Languages, numerical content, formatting, and code can create further differences.

The inference cost can be much higher than the output sticker price. For the \$10 input/\$50 output tariff, an input/visible-output ratio of four and ten additional reasoning tokens per visible token

yield $590 per million native visible tokens. A native/reference ratio of two and a 50% acceptance probability raise that scenario to $2,360 per million reference tokens. This is an explicit stress calculation, not observed usage or a bound on costs. Longer contexts, higher reasoning budgets, tool charges, or harder tasks could raise costs further.

6. Sensitivity, review, and task quality

6.1 Human-rate sensitivity

The available data support several checks on the task-rate calculation.

Table 4. Alternative sample and time definitions

Sample or time definition	N	Tokens/hour
Main: four occupations; reported task time	149	860
Broader sample including managers	278	900
Four occupations; explicit first-task no ChatGPT	55	796
Four occupations; no large paste event	113	838
Four occupations; no AI-use exclusions	165	889
Four occupations; fixed-time arm included	189	905
Four occupations; control second task, reports no ChatGPT	62	898
Main sample; browser page time instead	149	993

Notes: Each row is a separate descriptive sample or denominator choice. Self-reported task time is the main denominator; browser page time is a sensitivity. AI-response-based samples are selected and do not validate an unobserved population of unaided workers. Sample sizes and definitions are generated from the saved analysis output.

The larger 278-person sample, including managers, produces approximately 900 tokens per task hour, compared with 860 in the narrow sample. Removing large paste events and using explicit self-reported non-use provide checks on how reported AI use and recording behavior affect the estimates. These checks cannot establish the absence of hidden assistance. Browser page duration can omit offline preparation or reflect navigation behavior, while self-reports can be inaccurate. Neither should be treated as a gold-standard time clock.

A worker-level nonparametric bootstrap with 4,000 resamples gives a conditional 95% interval of about $47,852–$56,611 for the narrow sample's full-utilization wage calibration, holding the wage crosswalk and measurement choices fixed. This interval measures within-sample resampling variability. It excludes wage-survey uncertainty, convenience-sample selection, unknown AI use, reporting error, utilization uncertainty, and the mismatch of populations and dates. It is not a confidence interval for the average U.S. worker's cost.

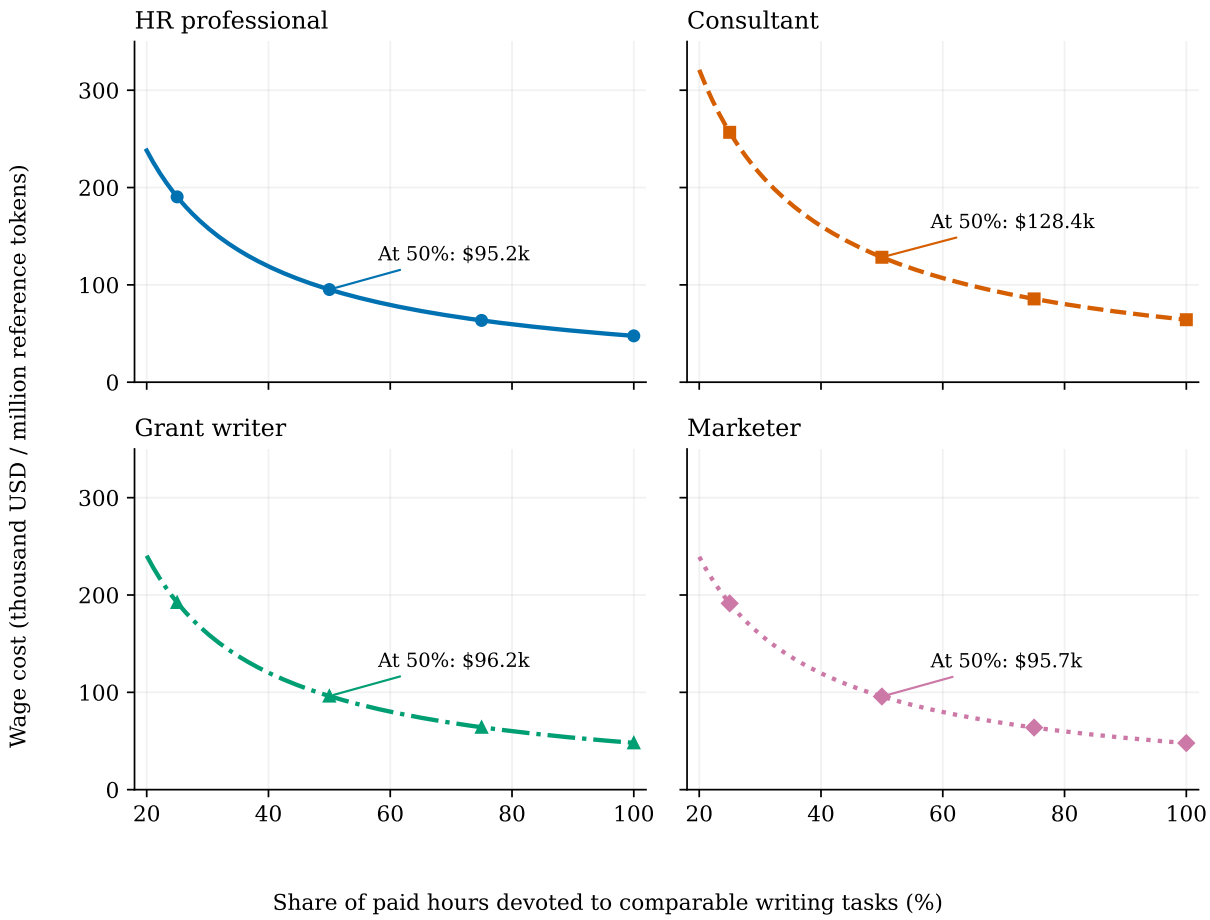


Figure 2: Human wage cost rises inversely with the assumed share of paid time devoted to comparable writing tasks. Each occupation is shown in a separate panel with identical axes because the HR, grant-writer, and marketer curves nearly overlap. Labels give the cost at 50% utilization; utilization is assumed, not observed in the experiment.

As an additional wage sensitivity, matching the narrow sample to its own reported annual salary and typical weekly hours gives approximately 37,163 survey dollars per million reference tokens for 146 observations, using salary divided by (52 times weekly hours). Annual reports above 500,000, nonpositive values, and weekly hours above 100 are excluded; excluding annual reports below 10,000 gives 40,110 for 132 observations. These figures are not substituted for the U.S. wage calibration because the archived materials do not establish a common geographic or currency normalization, year-round paid weeks are assumed, and no wage-date adjustment is made. They are an appendix-level check on the dependence on the BLS wage crosswalk.

6.2 Review can dominate generation

For a 1,000-reference-token document, the full-utilization human calibration is about $52.04 and the half-utilization calibration about $104.08. The high-price API benchmark is $0.06, before additional reasoning, retries, or review; the stress scenario above is $2.36. At a reviewer cost of $50

per paid hour, 30 minutes of review adds $25.00. The resulting model-assisted charges are $25.06 and $27.36 per 1,000 reference tokens, respectively, before other costs.

The corresponding accounting break-even review allowance is

$$m^* = \frac{60}{c_R}\left(\frac{H(u) - M_R}{1000}\right),$$

when $H(u) > M_R$. At $50 per hour and full writer utilization, it is approximately 62.4 minutes per 1,000-reference-token document for the API benchmark, or 59.6 minutes for the stress scenario. At half writer utilization, the benchmark allowance is approximately 124.8 minutes. These numbers do not show that a reviewer can detect all errors within that time. They say how much paid handling time would exhaust the modeled price advantage if quality and the remaining work were otherwise comparable.

Writer and reviewer time must be treated consistently. If the stated review minutes measure only active inspection, a separate utilization or overhead adjustment may be needed for the reviewer. Conversely, if organizational preparation and review require the same human work as original production, cheap generation cannot remove that cost. Expected losses from erroneous advice or defective work can exceed generation and review charges and are not estimated here.

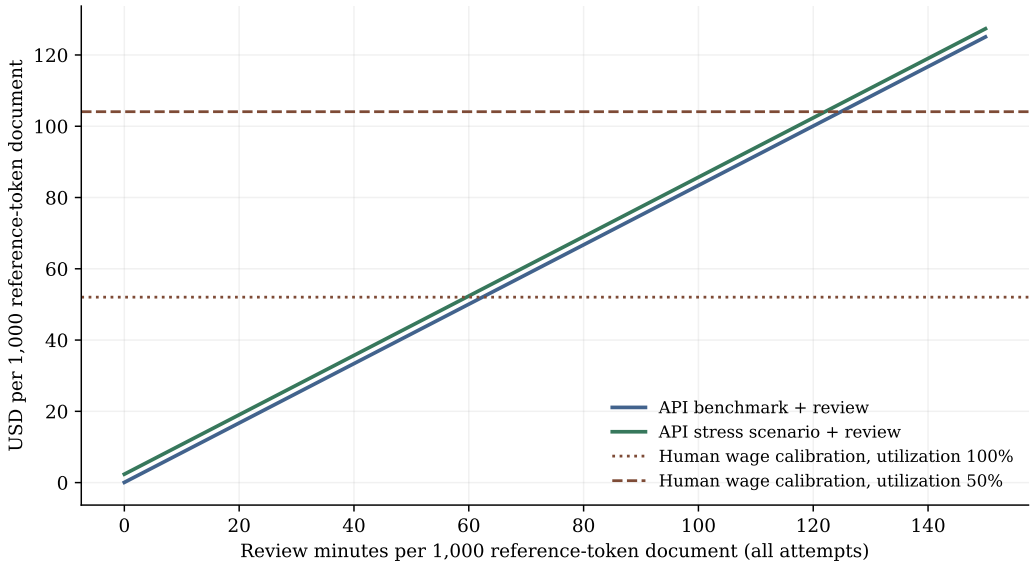


Figure 3: Review costs at $50 per paid hour, compared with the pooled human calibration. API benchmark: input/output = 1, no additional reasoning, native/reference = 1, acceptance = 1. Stress scenario: input/output = 4, reasoning/visible = 10, native/reference = 2, acceptance = 0.5.

6.3 Why this is not a substitution estimate

There is direct evidence that AI access changes task productivity, but it answers a different question. Noy and Zhang's randomized experiment reports a 40% reduction in task time and an 18%

improvement in graded output quality. Those causal findings concern a human using GPT-3.5 on the studied tasks, not the independent replacement of the worker by a current model. The authors emphasize that the assignments favored clear, relatively generic writing and did not require context-specific knowledge or precise factual accuracy (Noy and Zhang 2023a).

Brynjolfsson, Li, and Raymond's November 2023 working-paper version studies 5,179 customer-support agents and reports a 14% increase in issues resolved per hour, with a 34% gain among novice and lower-skilled workers. Its sample was predominantly outside the United States, and its outcome was successful resolutions rather than token production. These figures are cited specifically to that working-paper version, not silently assigned to the later journal publication (Brynjolfsson, Li, and Raymond 2023).

Together, these studies motivate the importance of task outcomes and human-AI complementarity. They do not justify converting the raw-token price gap into an estimate of employment losses, wage changes, or welfare gains. Longer responses can be less useful than shorter ones; deleting text may improve a document. A cost-per-token measure should therefore supplement, rather than replace, dollars per accepted task and task-specific quality measures.

7. Limitations and a better empirical design

Five limitations constrain the inference. First, the experiment is a convenience sample of short assignments from 2023; it contains neither a nationally representative work diary nor annual accepted-output records. Second, the BLS wage crosswalk assigns occupational means to participants whose actual compensation and geography are not matched. Third, self-reported task time and imperfect AI-use information introduce measurement uncertainty. Fourth, model tariffs are prices of a service, not estimates of its underlying resource cost, and they are reported in provider-specific token units. Fifth, contemporary models were not run on these same tasks and scored against a common acceptance standard.

A stronger design would recruit workers from specified occupations and observe a representative sample of paid days. For each task it would record the brief, input materials, all human time, revisions, final accepted output, compensation, and relevant task complexity. Human-only and model-assisted workflows would be assigned to comparable tasks, with independent evaluators applying a prespecified quality rubric. Model runs would record actual billed input, visible output, hidden reasoning, tools, cache state, failed attempts, and review time. Reported outcomes would include cost per accepted document and task-success rates, with common-token counts as a secondary normalization.

Such a design would also distinguish productivity effects from changes in demand. If cheaper writing induces more documents, additional checking, or more ambitious work, the effect on labor demand may differ from the effect on unit cost. Conversely, a task with high verification costs may benefit less than its generation price suggests. Those mechanisms are outside the identification supplied by this descriptive calibration.

8. Conclusion

For a selected set of professional-writing tasks, experimentally observed output rates and U.S. wage proxies imply wage costs in the tens to hundreds of thousands of dollars per million reference tokens, depending chiefly on the share of paid time devoted to comparable writing. Current advertised model generation tariffs in the selected sample are orders of magnitude lower in their native billing units.

The useful economic result is that generating candidate text can be very inexpensive relative to compensated human production time. The unresolved question is the cost of producing acceptable work with the necessary information, judgment, review, and accountability. The paper makes those missing quantities explicit rather than assigning an unsupported universal cost per token to an "average human."

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Appendix A. Occupational selection and excluded shortcuts

The initial search considered technical writers, authors, medical transcriptionists, and court reporters because their work can produce countable text. Public wage data alone were insufficient. Technical-writing work often includes interviewing, testing, diagrams, and revision; without matched final-output and time data, a wage cannot identify a token rate. Medical transcription requires a definition of a line, editing effort, audio difficulty, and whether speech recognition is already used. No national mean production rate satisfying those conditions was established here.

Court reporting illustrates the danger of using a professional speed standard as an average. The NCRA Registered Professional Reporter skills test specifies five-minute dictations at 180, 200, and 225 words per minute, but its workflow also allows three minutes to attach steno notes after dictation, followed by 75 minutes to transcribe and submit the final transcript. Multiplying dictation speed by every paid hour would ignore that workflow and turn a certification threshold into a measured occupational mean. These standards are therefore discussed as evidence about measurement, not entered as the output of an average court reporter.

Software development, spoken conversation, meetings, reading, and editing of existing documents are excluded from the main output denominator. Code tokens do not measure delivered functionality; reviewed document tokens are not all newly produced text. These activities can be studied with task-specific outcomes, but pooling them with original prose would make the aggregate difficult to interpret.

The selected four occupations are not a claim that all their output is text. They are the subset of the available experiment for which submitted prose and a transparent wage proxy can be combined. Their population employment counts are not used to manufacture an economy-wide average from experimental task rates.

Appendix B. API identifiers and tariff qualifications

- **OpenAI:** `gpt-6-astra`, `gpt-6.1-sol`, and `gpt-6-luna`. Astra/Sol base tariffs apply at up to 272,000 input tokens; above that threshold, full-request input/cache rates double and output rates rise 50%. Batch rates are half the standard table. Reasoning is billed as output, including reasoning not exposed in the visible answer.
- **Anthropic:** `claude-fable-5-1`, `claude-opus-5-5`, `claude-sonnet-5-5`, and `claude-haiku-5-5`. Haiku's base rate applies up to 100,000 prompt tokens; above that it is $0.50 input/$2.50 output per million. The other listed models include the full 1M context at standard pricing. US-only routing adds 10%; batch halves input/output charges. Thinking is billed even when hidden.
- **Google:** `gemini-4-argon` and `gemini-3.8-flash`. Argon's listed $2/$10 input/output rates apply through November 30, 2026, then become $4/$20. Flash's $0.75/$3.75 rates apply through December 31, 2026, then become $1.50/$7.50. Output prices include thinking tokens. Scheduled increases are not silently treated as the collection-date price.
- **DeepSeek:** `deepseek-flash` mapped to DeepSeek-V4.1-Flash-1017, and `deepseek-v4-pro`

mapped to DeepSeek-V4-Pro-0813. Peak cache-miss input/output prices are $0.30/$1.20 and $1.32/$3.96. Off-peak rates are half those levels. The documented peak periods are 01:00–04:00 and 06:00–10:00 UTC on weekdays excluding Chinese public holidays. Peak is an explicit tariff selection, not a claim about every retrieval instant. Completion usage includes a reasoning-token breakdown. Flash's model README and Pro's model metadata identify MIT licensing.

- **Mistral:** `mistral-large-4`, public preview v26.10. The official model card identifies open weights and $1.36 input/$4.18 output per million. Exact license and any separate reasoning-billing rule were not established by this collection; an open-source licensing claim is therefore not made.

No paid inference was performed, no API keys were required, and no model was scored against the experimental submissions. The price comparison covers purchased hosted inference. A self-hosted comparison would require an explicit hardware choice, measured throughput, serving configuration, utilization, power and operations costs, and required availability. Those quantities are unmeasured here.

Appendix C. Reproducibility and source audit

The accompanying project contains the protocol, raw-source ledgers, pinned Python dependencies, numerical intermediates, figures, and paper source. The reference tokenizer is `cl100k_base` from `tiktoken==0.12.0`. Analysis seeds are fixed; the human-rate bootstrap uses seed 20261021 and the wage-calibration bootstrap uses 20261022. The paper's aggregate is reconstructed from the saved observation-level numeric table without publishing essay text in the paper.

The core workflow is:

```
bash scripts/setup.sh
bash scripts/build.sh
```

A full microdata rebuild requires the archived public raw files retained in this workspace. The downloadable reproducibility bundle contains numeric derived data, manifests, code, and the paper; full original essays and publisher PDFs are excluded from that convenience bundle. The source-restoration script can download required original artifacts and verify their recorded hashes. If a source has changed, it stops rather than silently replacing the historical snapshot.

`extract_wages.py` extracts specified SOC rows and compensation components from archived BLS sources. `analyze_human_experiment.py` reproduces the author sample, counts final text, applies documented filters, and exports the task measures and audit tables. `analyze_costs.py` applies explicit wage crosswalks, computes aggregate ratios, scenario grids, review thresholds, and figures. `render_paper.py` inserts generated tables into the paper template. `validate_results.py` checks accounting identities, primary sample counts, source hashes for inputs, and the relationship between reported totals and calculated rates. Pandoc and XeLaTeX generate the PDF.

Sources that redirect to generic landing pages are preserved as unsuccessful data attempts and are not treated as occupation observations. Model marketing pages that rejected automated requests

were replaced by successfully retrieved official developer documentation. Earlier and later versions of studies are not combined silently. All substantive model-price figures in Table 3 were checked against saved official tables; no remembered model tariff substitutes for a captured source.